



CS 260 – Foundations of Data Science

Lecture 9 – Laplacian Smoothing & Decision Trees

09/29/2026

Adam Poliak



Announcements

HW4 due ~~Tuesday~~ Friday night

Midterm 1 – October 8th

Office Hours:

Tuesdays: 4pm-4:30pm

This Thursday until 2:15pm



Outline

Naïve Bayes Algorithm

Decision Trees



Components of a Bayes Model

$$p(y = k|\mathbf{x}) = \frac{p(y = k)p(\mathbf{x}|y = k)}{p(\mathbf{x})}$$

Components of a Bayes Model

Identify the evidence, prior, posterior, and likelihood in the equation below

$$p(y = k|\mathbf{x}) = \frac{p(y = k)p(\mathbf{x}|y = k)}{p(\mathbf{x})}$$

Components of a Bayes Model

Identify the **evidence** prior, posterior, and likelihood in the equation below

$$p(y = k | \mathbf{x}) = \frac{p(y = k)p(\mathbf{x} | y = k)}{p(\mathbf{x})}$$

Evidence: this is the data (features) we observe, which we think will help us predict the outcome we're interested in



Components of a Bayes Model

Identify the evidence, prior, posterior, and likelihood in the equation below

$$p(y = k | \mathbf{x}) = \frac{p(y = k)p(\mathbf{x} | y = k)}{p(\mathbf{x})}$$

Prior: without seeing any evidence (data), what is our prior believe about each outcome (intuition: what is the outcome in the population as a whole?)



Components of a Bayes Model

Identify the evidence, prior, posterior, and likelihood in the equation below

$$p(y = k | \mathbf{x}) = \frac{p(y = k)p(\mathbf{x} | y = k)}{p(\mathbf{x})}$$

Posterior: this is the quantity we are actually interested in. **Given** the evidence, what is the probability of the outcome?



Components of a Bayes Model

Identify the evidence, prior, posterior, and likelihood in the equation below

$$p(y = k|\mathbf{x}) = \frac{p(y = k)p(\mathbf{x}|y = k)}{p(\mathbf{x})}$$

Likelihood: given an outcome, what is the probability of observing this set of features?



Naïve Bayes

$$\begin{aligned} p(\vec{x}|y = k) &= p(x_p|y = k)p(x_{p-1}|y = k) \dots p(x_2|y = k) p(x_1|y = k) \\ &= \prod_{j=1}^p p(x_j|y = k) \end{aligned}$$

Naïve Bayes Model

$$p(y = k|\vec{x}) \propto p(y = k) \prod_{j=1}^p p(x_j|y = k)$$

proportional to



Naive Bayes Example

	Cat	Documents
Training	-	just plain boring
	-	entirely predictable and lacks energy
	-	no surprises and very few laughs
	+	very powerful
	+	the most fun film of the summer

Test: powerful very fun

$$p(+) = ?$$

$$p(-) = ?$$



Naive Bayes Example

	Cat	Documents
Training	-	just plain boring
	-	entirely predictable and lacks energy
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	+	very powerful
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Test: predictable with no fun

$$p(+) = ?$$

$$p(-) = ?$$



Naive Bayes Example

	Cat	Documents
Training	-	just plain boring
	-	entirely predictable and lacks energy
	-	no surprises and very few laughs
	+	very powerful
	+	the most fun film of the summer

Test: entirely boring no fun

$$p(+) = ?$$

$$p(-) = ?$$



Laplacian Smoothing

$$\hat{P}(w_i|c) = \frac{\text{count}(w_i, c) + 1}{\sum_{w \in V} (\text{count}(w, c) + 1)} = \frac{\text{count}(w_i, c) + 1}{(\sum_{w \in V} \text{count}(w, c)) + |V|}$$

Naive Bayes Example

	Cat	Documents
Training	-	just plain boring
	-	entirely predictable and lacks energy
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	+	very powerful
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Test: predictable with no fun

$$p(+) = ?$$

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Outline

Naïve Bayes Algorithm

Decision Trees



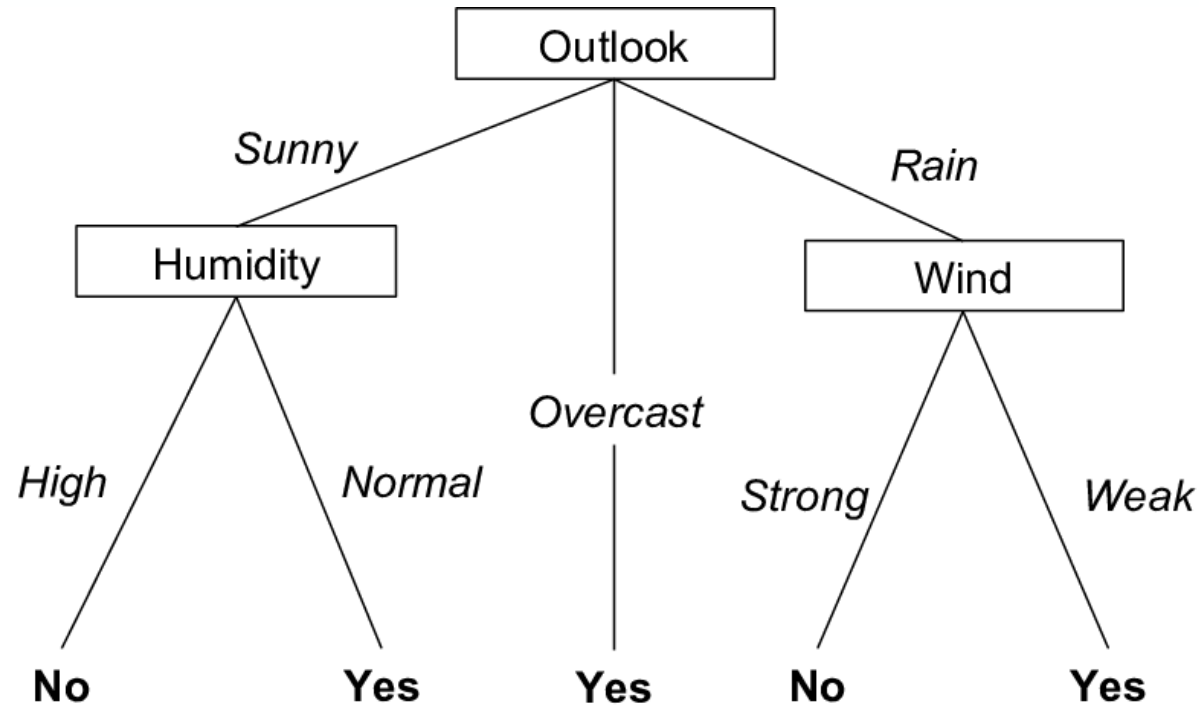
Tennis example

Day	Outlook	Temperature	Humidity	Wind	PlayTennis (y)
x_1	Sunny	Hot	High	Weak	No
x_2	Sunny	Hot	High	Strong	No
x_3	Overcast	Hot	High	Weak	Yes
x_4	Rain	Mild	High	Weak	Yes
x_5	Rain	Cool	Normal	Weak	Yes
x_6	Rain	Cool	Normal	Strong	No
x_7	Overcast	Cool	Normal	Strong	Yes
x_8	Sunny	Mild	High	Weak	No
x_9	Sunny	Cool	Normal	Weak	Yes
x_{10}	Rain	Mild	Normal	Weak	Yes

Data from Machine Learning by Tom Mitchell (Table 3.2)



Decision Tree example (tennis)



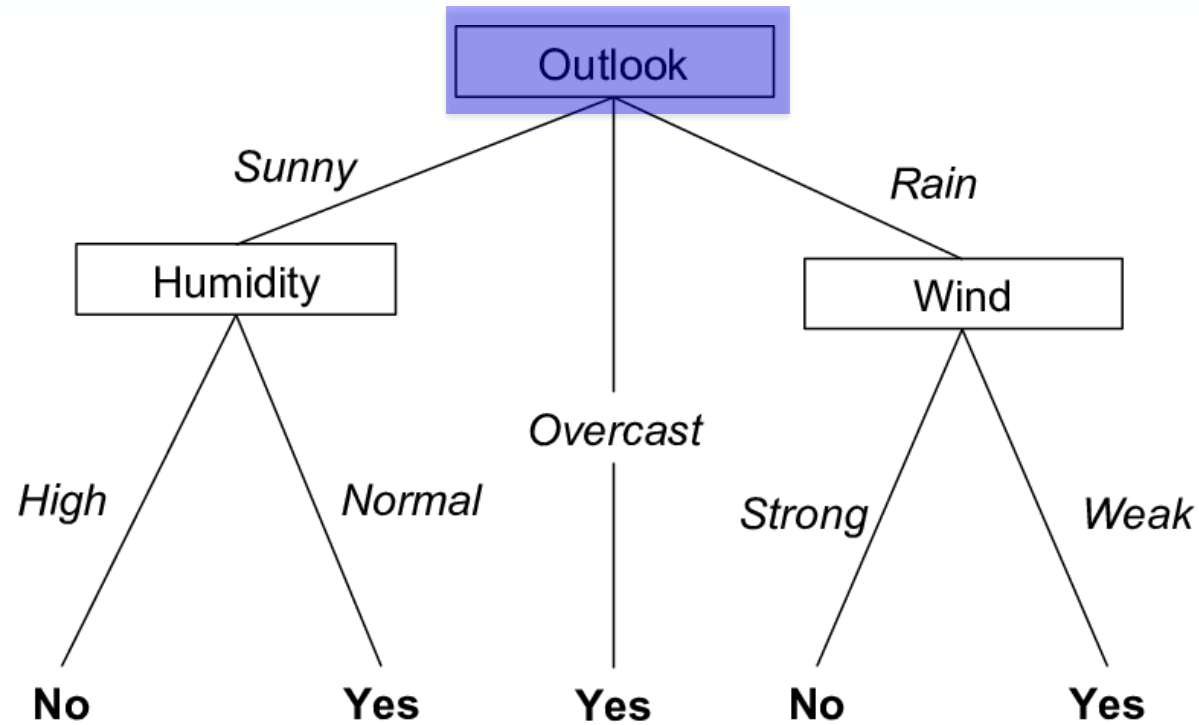
Each internal node: test one feature

Each branch from node: selects one value of the feature

Each leaf node: predict y



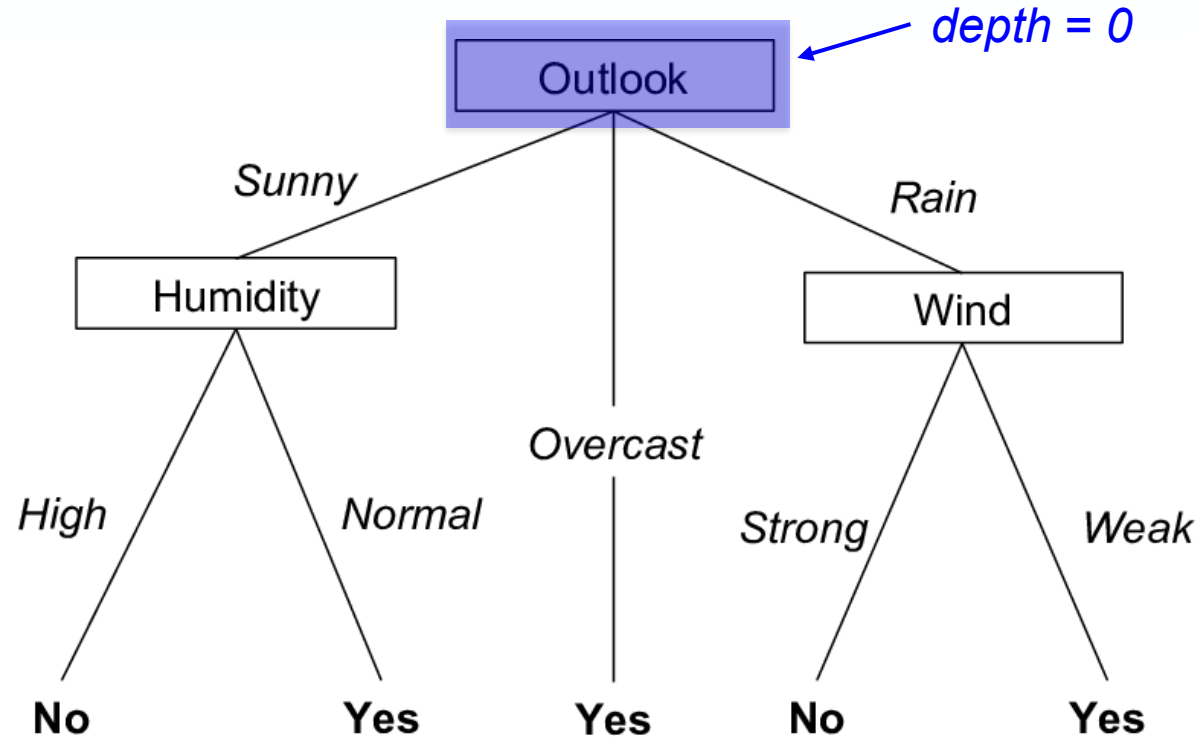
Decision Tree example (tennis)



Key term: *depth*



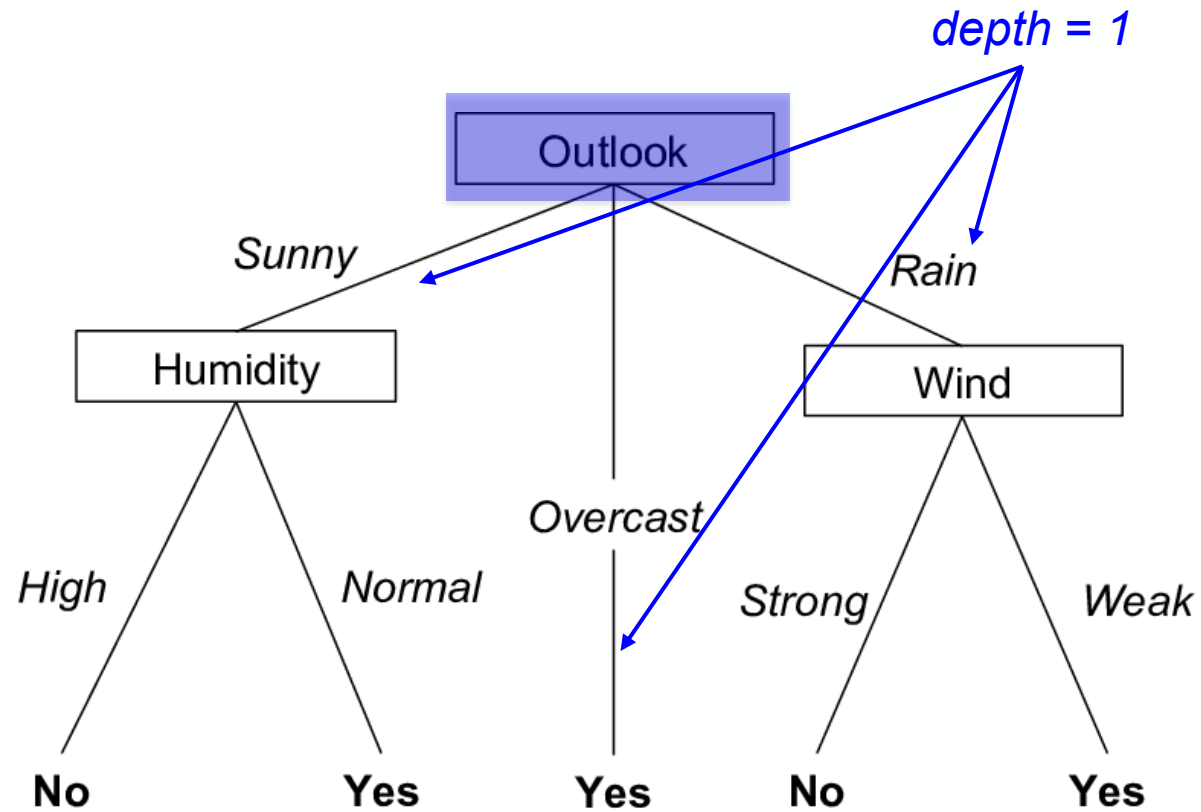
Decision Tree example (tennis)



Key term: *depth*



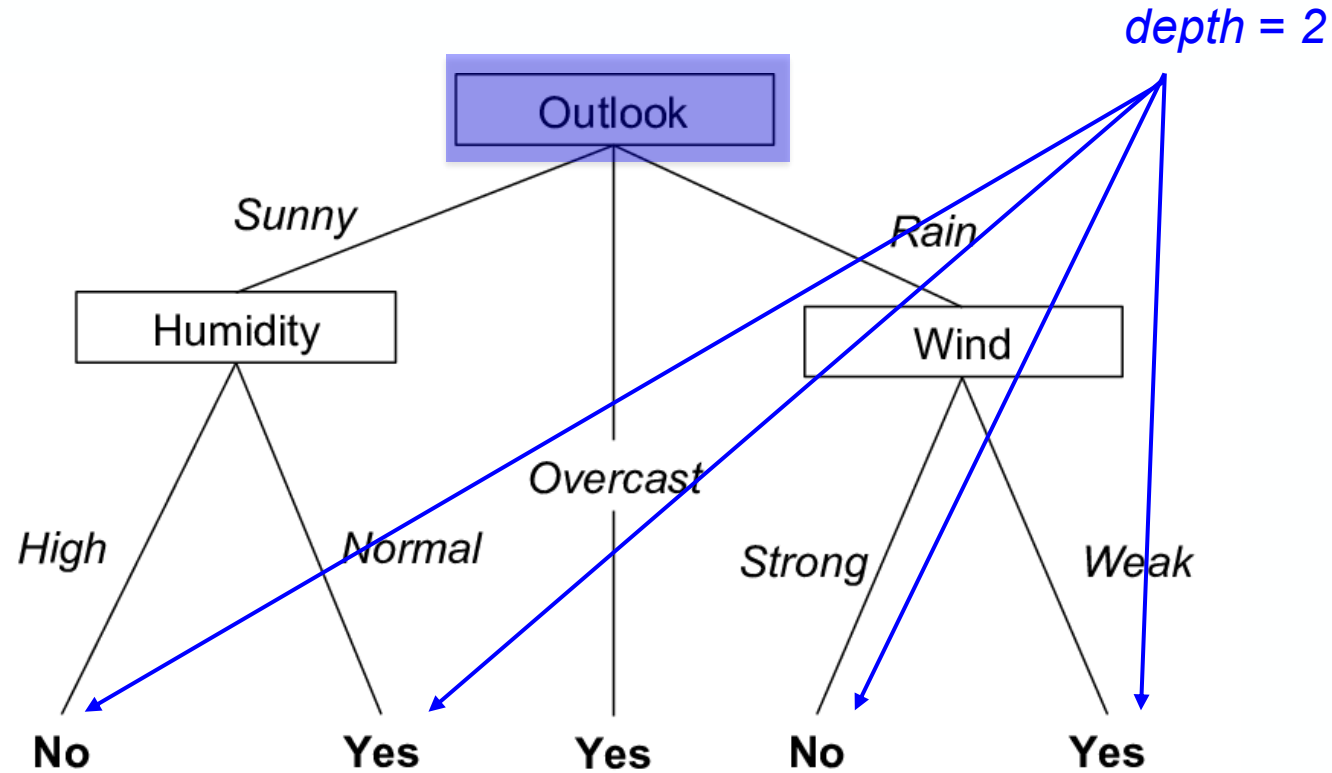
Decision Tree example (tennis)



Key term: *depth*



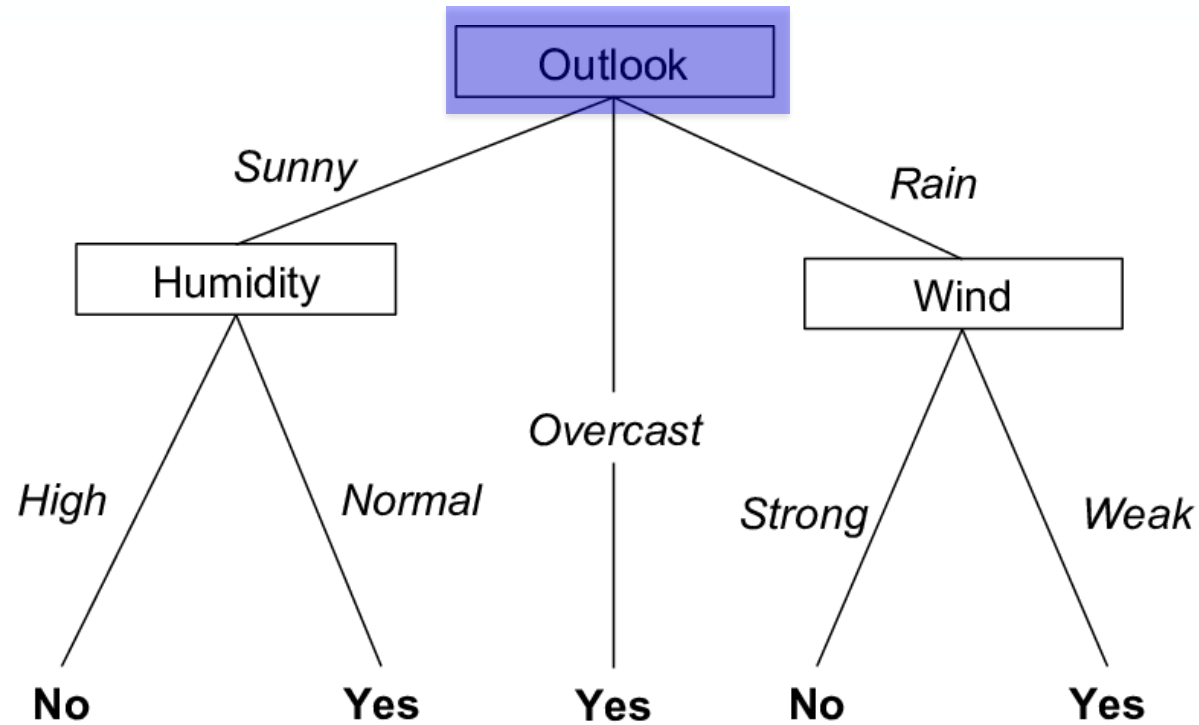
Decision Tree example (tennis)



Key term: *depth*



Decision Tree example (tennis)

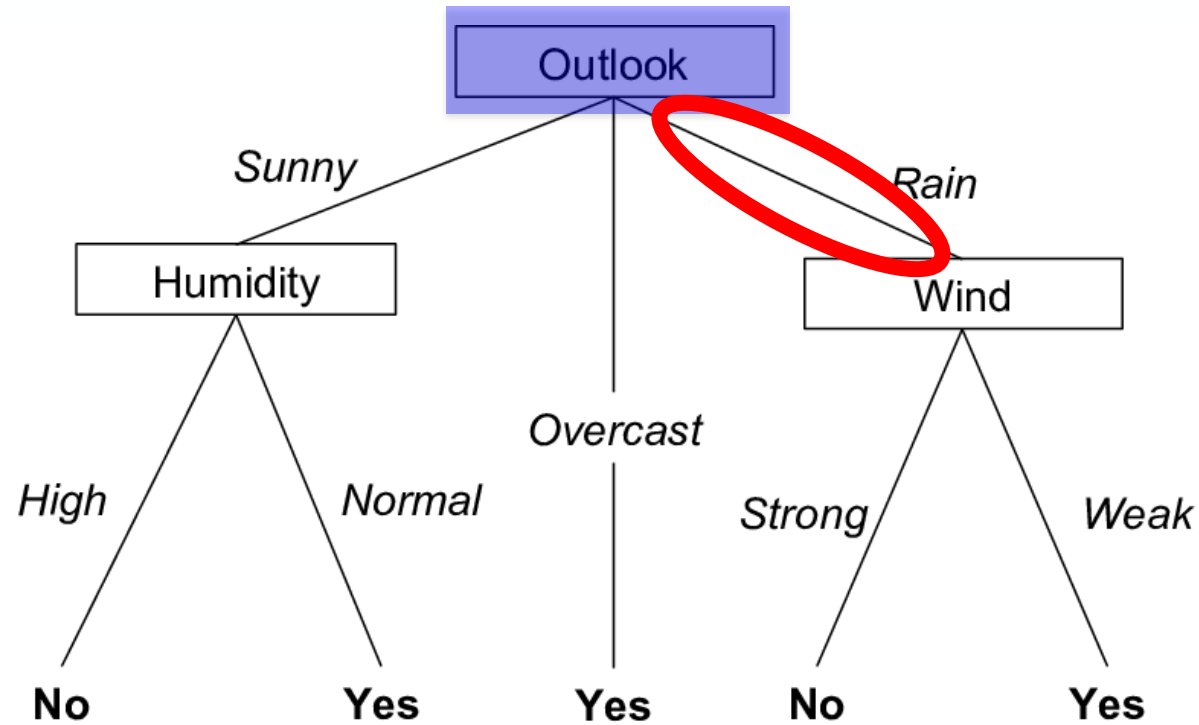


(test example) $\mathbf{x} =$

Outlook	Temp	Humidity	Wind
Rain	Hot	High	Strong



Decision Tree example (tennis)

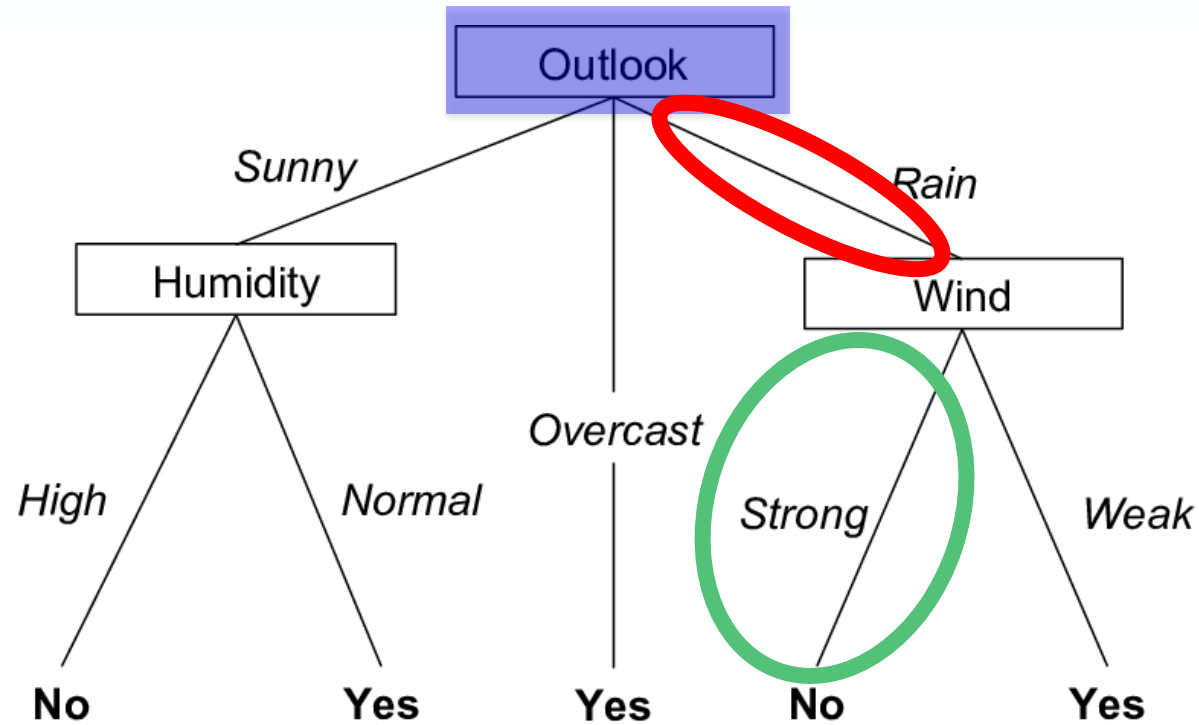


(test example) $\mathbf{x} =$

Outlook	Temp	Humidity	Wind
Rain	Hot	High	Strong



Decision Tree example (tennis)

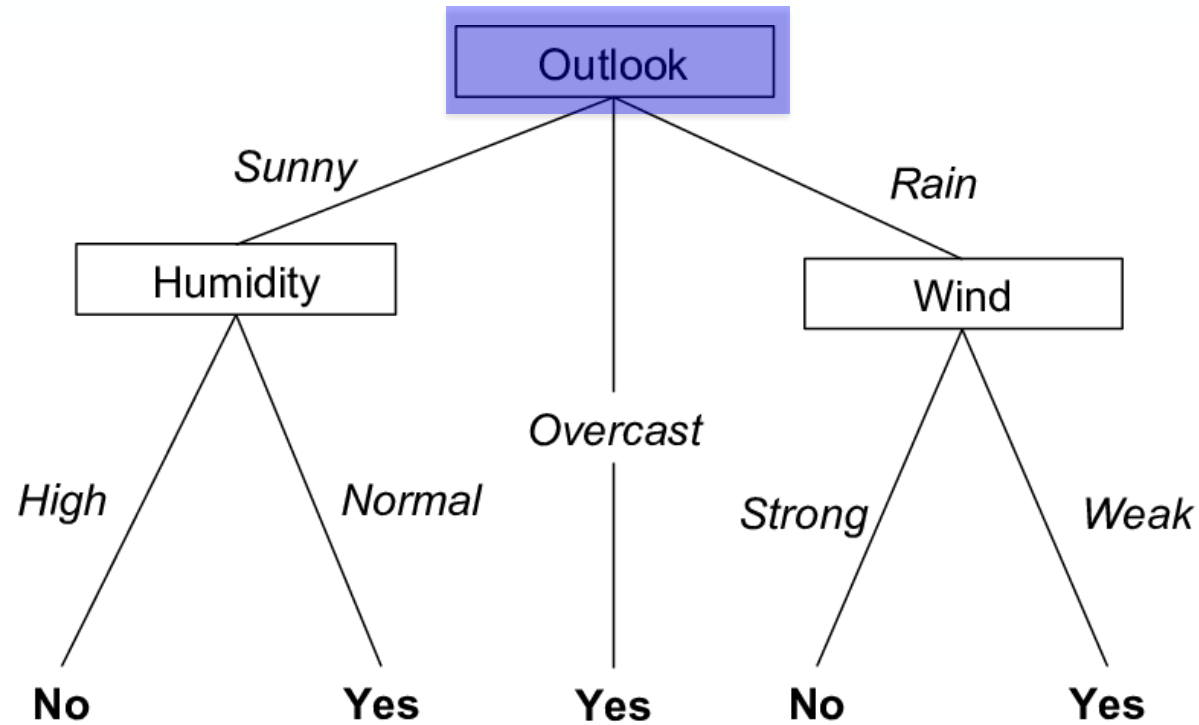


(test example) $\mathbf{x} =$

Outlook	Temp	Humidity	Wind
Rain	Hot	High	Strong



Decision Tree example (tennis)



(test example) $\mathbf{x} =$

Outlook	Temp	Humidity	Wind
Rain	Hot	High	Strong

$\hat{y} = \text{No}$



Decision Trees: Pros vs Cons

Discuss with a partner! Think about:

- * training and testing
- * featurization
- * runtime
- * human factors



Decision Trees: Pros vs Cons

- Very interpretable! Easy to say *why* we made a classification (can point to which features)
- Compact representation and fast predictions
- Can be brittle (not looking at each example holistically)
- Featurization and implementation difficulties



ID3 Decision Tree Algorithm

Select feature that “best” informs label prediction (i.e. y)

Divide: partition data into branches based on their value at this feature

Conquer: recurse on each partition

Posted as optional reading

Machine Learning 1: 81–106, 1986
© 1986 Kluwer Academic Publishers, Boston — Manufactured in The Netherlands

Induction of Decision Trees

J.R. QUINLAN (munari!nswitgould.oz!quinlan@seismo.css.gov)
Centre for Advanced Computing Sciences, New South Wales Institute of Technology, Sydney 2007, Australia

(Received August 1, 1985)

Key words: classification, induction, decision trees, information theory, knowledge acquisition, expert systems



Top-Down Decision Tree Algorithm

$\text{MakeSubtree}(D, F)$



Top-Down Decision Tree Algorithm

MakeSubtree(D, F)

- if stopping criteria met

 - make a leaf node N

 - determine class label/probabilities for N



Top-Down Decision Tree Algorithm

$\text{Dataset } (X,y)$

$\text{MakeSubtree}(D, F)$

- if stopping criteria met
- make a leaf node N
- determine class label/probabilities for N



Top-Down Decision Tree Algorithm

$\text{MakeSubtree}(D, F)$

Dataset (X,y)

Features

if stopping criteria met

make a leaf node N

determine class label/probabilities for N



Top-Down Decision Tree Algorithm

$\text{MakeSubtree}(D, F)$

Dataset (X,y) → D
Features → F

- if stopping criteria met
 - make a leaf node N
 - determine class label/probabilities for N
- else
 - make an internal node N
 - $S = \text{FindBestFeature}(D, F)$
 - for each outcome k of S
 - $D_k =$ subset of instances that have outcome k
 - $N.\text{child}[k] = \text{MakeSubtree}(D_k, F-S)$

return subtree rooted at N



Top-Down Decision Tree Algorithm

$\text{MakeSubtree}(D, F)$

Dataset (X,y) → D

Features → F

if stopping criteria met

make a leaf node N

determine class label/probabilities for N

else

make an internal node N

$S = \text{FindBestFeature}(D, F)$

for each outcome k of S

D_k = subset of instances that have outcome k

$N.\text{child}[k] = \text{MakeSubtree}(D_k, F-S)$

return subtree rooted at N

Why don't we want to use this feature again?



Design choice: stopping criteria

1. All the data points in our partition have the same label
2. No more features remain to split on
3. No features are informative about the label
4. Reached (user specified) max depth in the tree

How to select “best” features?

X			Y
Color	Shape	Size	Likes toy?
red	square	big	+
blue	square	big	+
red	circle	small	-
blue	square	small	-
red	circle	big	+



How to select “best” features?

X			Y
Color	Shape	Size	Likes toy?
red	square	big	+
blue	square	big	+
red	circle	big	-
blue	square	big	-
red	circle	big	+



Most Informative Feature

Goal: Which feature will allow us to learn the most information

Information Gain – metric from information theory

“the amount of information gained about a random variable or signal from observing another random variable”

Wikipedia

Entropy

measure of uncertainty in a group of observations



Which feature has the highest IG

X			Y
Color	Shape	Size	Likes toy?
red	square	big	+
blue	square	big	+
red	circle	small	-
blue	square	small	-
red	circle	big	+



Size and entropy

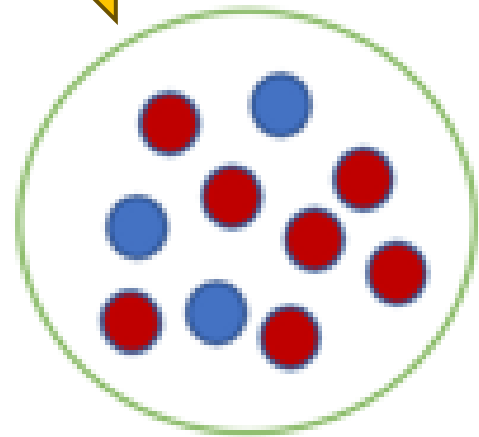
X			Y
Color	Shape	Size	Likes toy?
red	square	big	+
blue	square	big	+
red	circle	small	-
blue	square	small	-
red	circle	big	+



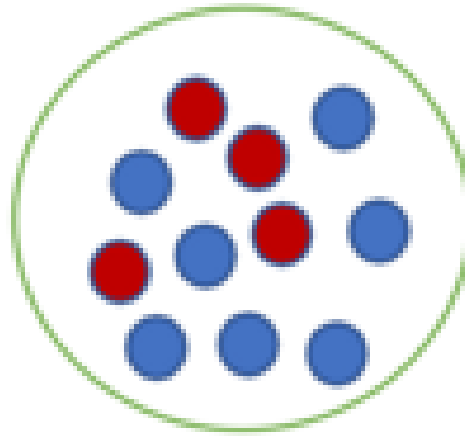
Entropy & Information Gain

Low IG
High Entropy

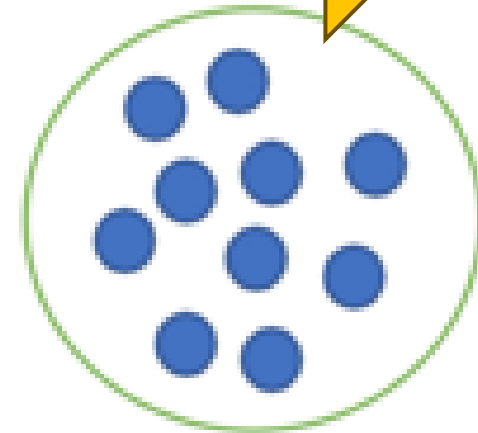
High IG
Low Entropy



$$p(\text{red}) = .7$$
$$p(\text{blue}) = .3$$



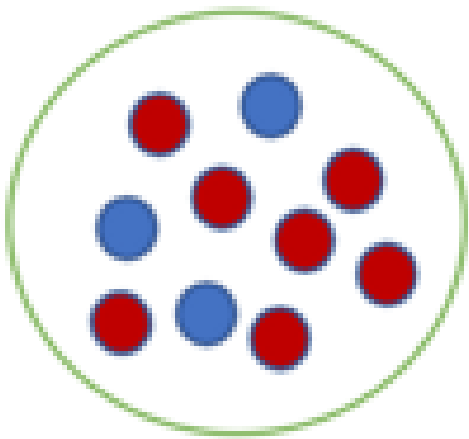
$$p(\text{red}) = .4$$
$$p(\text{blue}) = .6$$



$$p(\text{red}) = 0$$
$$p(\text{blue}) = 1$$

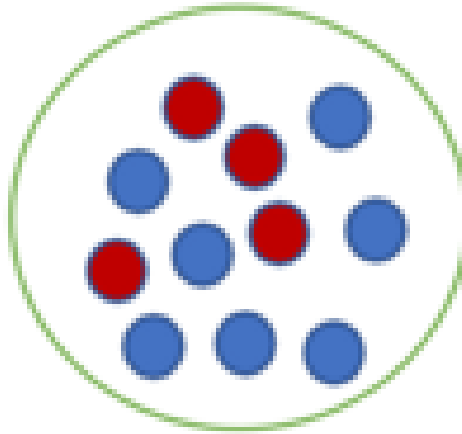
Quantify Entropy

$$\text{Entropy: } \sum_i^c p_i \log(p_i)$$



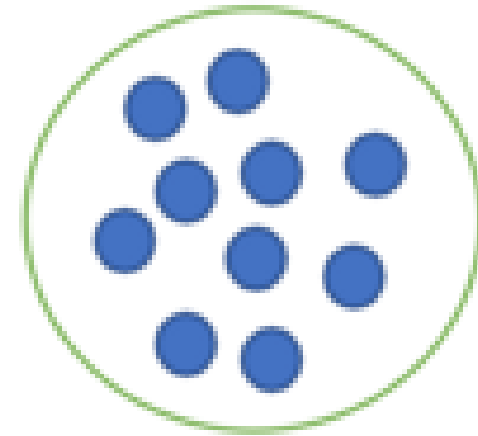
$$p(\text{red}) = .7$$

$$p(\text{blue}) = .3$$



$$p(\text{red}) = .4$$

$$p(\text{blue}) = .6$$

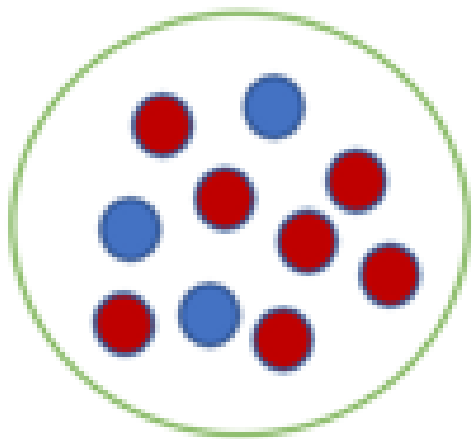


$$p(\text{red}) = 0$$

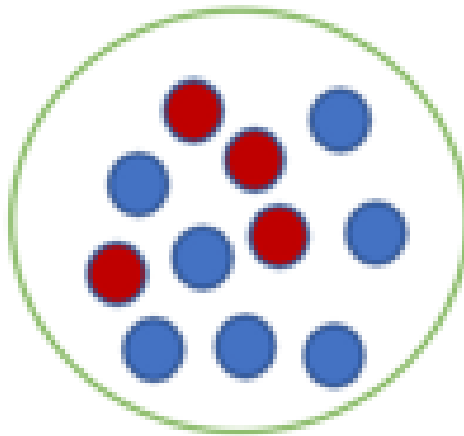
$$p(\text{blue}) = 1$$

Quantify Entropy

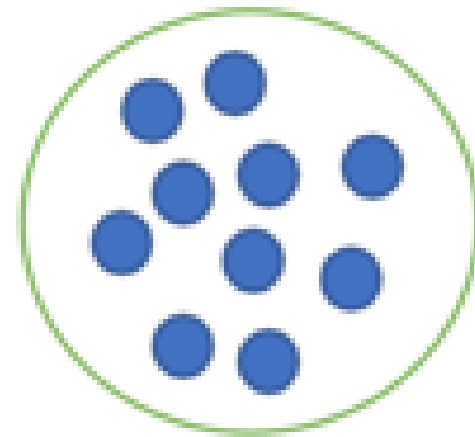
$$\text{Entropy: } \sum_i^c p_i \log(p_i)$$



-0.265



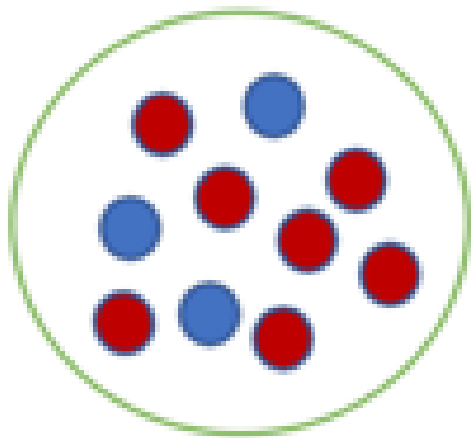
-0.292



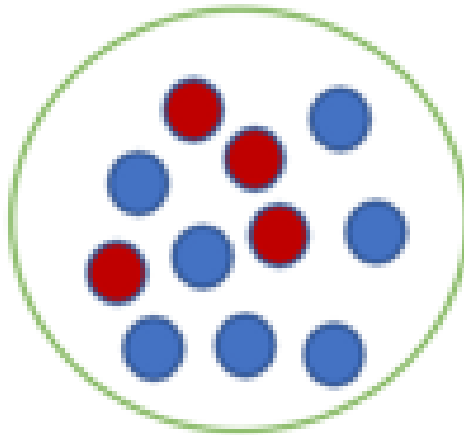
0

Quantify Entropy

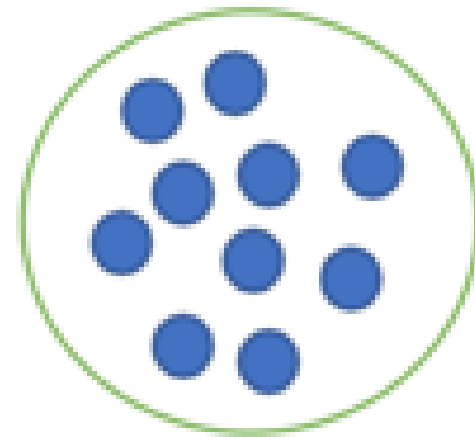
$$\text{Entropy: } -\sum_i^c p_i \log(p_i)$$



0.265



0.292



0

Conditional Entropy

X			Y
Color	Shape	Size	Likes toy?
red	square	big	+
blue	square	big	+
red	circle	small	-
blue	square	small	-
red	circle	big	+



Conditional Entropy

X			Y
Color	Shape	Size	Likes toy?
red	square	big	+
blue	square	big	+
red	circle	small	-
blue	square	small	-
red	circle	big	+

$$H(Y | X = v) = - \sum_i^c p(Y_i | X = v) \log(p(Y_i | X = v))$$



Conditional Entropy

$$H(Y | X = v): \sum_i^C p(y_i | X = v) \log(p(y_i | X = v))$$

$$H(Y | X): \sum_{v \in \text{dom}(X)}^{|X|} p(X = v) H(Y | X = v)$$

X			Y
Color	Shape	Size	Likes toy?
red	square	big	+
blue	square	big	+
red	circle	small	-
blue	square	small	-
red	circle	big	+



Information Gain

$$H(Y | X = v) = - \sum_{i=1}^c p(y_i | X = v) \log(p(y_i | X = v))$$

$$H(Y | X) = - \sum_{v \in \text{dom}(X)} p(X = v) H(Y | X = v)$$

$$IG(Y, X) = H(Y) - H(Y | X)$$

X			Y
Color	Shape	Size	Likes toy?
red	square	big	+
blue	square	big	+
red	circle	small	-
blue	square	small	-
red	circle	big	+

